BIOS 998-002

Ecosystem Modeling

Spring 2027

1 Course Description

Students in this course will learn the core concepts behind systems thinking and model building. Through course-based model development, students will learn the skills necessary to develop their own model based on their interests and to understand the workings of models developed by others. Models will be developed independently and connected collaboratively into a class-scale ecosystem model. The course is intended for graduate students interested in developing skills in quantitative theory development, systems thinking, and model building. Students will also learn best practices in R coding, version control, and large project collaboration.

1.1 Class Time and Location

Tuesdays and Thursdays 12:30-13:45

Beadle N172

1.2 Instructor

Dr. Nick Smith

Manter 402C

nsmith87@unl.edu

1.3 Office Hours

Thursdays 14:00-14:45

Manter 402C

1.4 Recommended Texts

Climate Change and Terrestrial Ecosystem Modeling by Bonan https://doi.org/10.1017/9781107339217

Principles of Terrestrial Ecosystem Ecology (2nd Edition; 2011) by Chapin, Matson, and Vitousek

The book can be accessed from Springer here: https://link.springer.com/book/10.1007/978-1-4419-9504-9. Click on "Access this title on SpringerLink." It can also be accessed through the TTU library.

2 Course Materials

All course materials, including lecture slides, readings, activities, and code will be posted to a GitHub repository for the course. The primary repository address is https://github.com/SmithEcophysLab/ecosystem_modeling_sp2027. The repository will include the syllabus and all other miscellaneous class materials as the semester progresses. A README file will contain information on the repository, including links to different sections at https://github.com/SmithEcophysLab/ecosystem_modeling_sp2027/README.md.

3 Learning Objective

This course will broadly focus on understanding the principles underlying the development of process-based models of ecosystems. An emphasis will be placed on how to build these models to understand the interactions between ecosystems and drivers of global change, including climate change, changes in atmospheric gas concentration, and eutrophication. Course topics will be taught through the lens of classical and more contemporary primary and secondary literature. There will be a strong emphasis on hands-on modeling skill development. Topics will be flexible and modified to match student interests where possible.

4 Attendance Policy

Attendance will not be taken, but is strongly recommended. In class activity points will only be granted if students are in class. Makeups will not be granted.

5 Course Assessment

5.1 *Participation and Engagement*

Being an active and engaged participant in the class will benefit your understanding of material as well as your peers'. Examples include asking questions, providing feedback, and facilitating discussion.

5.2 *Quizzes*

Short quizzes will be given periodically to test student knowledge of core concepts and to stimulate discussion. In some cases, quizzes will be developed and administered by class weekly co-leads.

5.3 *Homework Assignments*

Short homework assignments will be given each week based on the Tuesday lectures and Thursday model development discussions. These will typically be assigned on Thursday and due the following Tuesday.

5.4 *Weekly co-leads*

Throughout the semester, students will be asked to co-lead on the week's discussion topic with Dr. Smith. This will consist of leading a jigsaw-style discussion of a literature article on Tuesdays and developing a quiz for students on Thursdays. Students will be evaluated on their ability to respond accurately to their peers' questions as well as their ability to summarize and generate discussion on the week's topic.

5.5 *Model module development*

The primary semester project will be to develop a module for the class ecosystem model based on the student's interest. The module will be written in R and must be able to run as a stand-alone module as well as in connection with the larger class model. The student will be required to write a full description of the module to coincide with the code in both README and manuscript style format.

6 Grading

Participation and Engagement: 25%

Quizzes: 5%

Homework Assignments: 20%

Weekly co-lead: 15%

Module proposal: 10%

Module presentation: 5%

Final module: 20%

Grades will be made available on Blackboard. All grades posted at the end of the course will be final, unless an error has been made in their calculation. Please contact Dr. Smith if you feel your grade has been calculated incorrectly.

7 Grading Scale

A: $\geq$ 90%

B: 80 – 90%

C: 70 – 80%

D: 60 – 70%

F: $\leq$ 59.9%

8 Missing In-class Activities

Students will be required to be in class to receive in-class activity points. Please contact Dr. Smith if you plan to miss class for a university function *prior to class*. If class is missed due to an illness, please let Dr. Smith know as soon as possible. Documentation will need to be provided in order to be able to make up any missed work.

Schedule of Topics by Week

01/12/2027 – Introductions, semester planning, and goals

01/19/2027 – Systems Thinking; building systems diagrams

01/26/2027 – Terrestrial Ecosystem cycles; downloading and using R

02/02/2027 – Ecosystem Models; GitHub and the class model environment

02/09/2027 – 3 R's of Ecosystem Modeling; working collaboratively in GitHub

02/16/2027 – Integrating Ecology into Models; writing functions in R

02/23/2027 – Eco-evolutionary Optimality Theory; model repository structure

03/02/2027 – **Module proposal presentations**

03/09/2027 – **Module proposal presentations**

03/16/2027 – NO CLASS; SPRING BREAK

03/23/2027 – Model development: Carbon

03/30/2027 – Model Development: Water

04/06/2027 - TBD

04/13/2027 – Model Development: Nutrients

04/20/2027 – **Module presentations**

04/27/2027 – **Module presentations**; **Final module due**

General Weekly Schedule

Generally, each Tuesday will consist of a lecture and jigsaw-style paper discussion. Thursdays will consist of a quiz, paper discussion, and class model development exercise.